

1. What is a Plant? + Importance to Humans

Typical plant features

- Green, chlorophyll, photosynthesis.
- Mostly immobile, rooted.
- Cell wall with cellulose.
- Indeterminate growth (continued cell division).
- No nervous or excretory systems.
- Exceptions: parasitic plants (dodder, Indian pipe) lack chlorophyll; insectivorous plants trap insects.

Why plants matter

Plants are the base of all food chains and key to Earth's climate.

Main benefits (the "F's"):

- Food – crops, fruits, vegetables; spices; beverages (coffee, tea, cocoa); fermentation (bread, beer, wine).
- Fiber/Fabrication – wood, paper, cotton, flax, hemp; many synthetics from plant derived fossil fuels.
- Fuel – wood, coal, oil, natural gas (ancient plants); modern biofuels (ethanol, biodiesel).
- Pharmaceuticals – e.g. digitalis (foxglove), many drugs + traditional medicines.
- Fermentation – yeasts for bread, alcohol, vinegar.
- Flowers/Forests/Fancification & Fragrance – aesthetics, shade, windbreaks, habitat, perfumes.

Quick check: in your own words, why can't animals survive without green plants?

2. Major Plant Groups

Overview

- Angiosperms (flowering plants)
- Gymnosperms (conifers)
- Pteridophytes – ferns, horsetails, club mosses.
- Bryophytes – mosses, liverworts.
- Algae
- Fungi
- Lichens

Angiosperms

Flowering plants with flowers + fruits (seeds enclosed).

Two main groups:

- Monocots
 - o 1 cotyledon
 - o Leaves: parallel venation
 - o Floral parts: multiples of 3
 - o Usually no wood
 - o Examples: grasses, corn, lilies, orchids, bananas, palms.
- Dicots
 - o 2 cotyledons
 - o Leaves: net/reticulate venation
 - o Floral parts: mostly 4s or 5s

angiosperms

Derived from the Greek words (angeion), meaning "vessel" or "container," and (sperma), meaning "seed." The term literally means "contained seed," referring to the defining feature that their seeds are enclosed within an ovary (which develops into a fruit).

- o Many woody species
- o Examples: roses, carrots, elms, magnolias, apples, sunflowers.

Gymnosperm (Conifers)

- "Naked" seeds (on cones, not in fruits).
- Woody trees/shrubs; many evergreen; needles or scale leaves.
- Often aromatic resins.
- Thrive in cool temperate zones.
- Examples: pine, spruce, fir, juniper, arborvitae, yew, cypress, redwood, ginkgo.

Ferns, Horsetails, Club Mosses (Pteridophytes)

- True roots, stems, leaves, but no flowers, no seeds.
- Reproduce by spores.
- Many with underground rhizomes.
- Ferns: large compound leaves; moist shaded habitats.
- Horsetails: jointed, silica rich stems ("scouring rushes").
- Club mosses: small evergreen herbs with tiny leaves; moist woods, bogs.

Mosses and Liverworts (Bryophytes)

- Small plants; no true roots, stems, leaves.
- Reproduce by spores; prefer moist or harsh habitats.
- Mosses: stem like body with leafy scales, capsules with spores (e.g. peat moss).
- Liverworts:
 - o Thallose (flat ribbons on wet surfaces)
 - o Or leafy (small stems with leaf like scales).

Algae

- From unicellular to giant kelps.
- Pigments □ green, red, brown, yellow, etc.
- Mostly aquatic; vital base of food chains.
- Green algae: freshwater, colonies, important fish food; blooms can pollute water.
- Diatoms: "grass of the sea"; silica walls □ diatomaceous earth (abrasives, filters).
- Blue green algae (cyanobacteria): can fix nitrogen.
- Brown algae: giant kelps; food, animal feed, and source of gels (ice cream, cosmetics).
- Red algae: mostly marine.

Fungi

gymnosperm

Derived from the Greek words (gymnós), meaning "naked" or "uncovered," and (sperma), meaning "seed." The term literally means "naked seed," referring to the characteristic that their seeds are not enclosed within an ovary (fruit).

pteridophytes

The word Pteridophytes is a Neo-Latin compound term created from two Ancient Greek roots: Pterid- (from pterís, the fern) which ultimately comes from (pterón), meaning "feather" or "wing." -phyte (from phutón), meaning "plant." This literally translates to "feather-plant" or "fern-plant," describing the characteristic feather-like (or pinnate) appearance of the fronds found on most ferns, which are the most common members of this group.

bryophytes

It is derived from Ancient Greek roots: Bryo- (from brúon), meaning "tree moss" or "liverwort". -phyte (from phutón), meaning "plant". Therefore, the term literally means "moss plant" or "moss-like plant," referring to the group of non-vascular plants that includes mosses, liverworts, and hornworts.

- No chlorophyll, roots, stems, leaves, flowers.
 - Now in their own kingdom.
 - Filamentous hyphae; reproduce by spores.
 - Roles: decomposers (saprophytes) and parasites (diseases).
 - Visible fungi: mushrooms, toadstools, brackets, puffballs, morels, truffles.
 - Many microscopic: molds, yeasts, mildews, rusts, smuts.
 - Importance: food (mushrooms, cheeses), antibiotics (penicillin, streptomycin), fermentation.
- “Mushroom” vs “toadstool”: informal; edible vs poisonous, no technical difference.

Lichen

- Symbiosis of green/blue green alga + fungus.
- Alga: photosynthesis (food).
- Fungus: structure, protection, water and minerals.
- Grow on rocks, trees, soil, harsh climates; sensitive to air pollution (bioindicators).
- Do not harm host trees.

Checkpoint: can you list two differences between angiosperms and gymnosperms?

3. Plant Classification & Naming

Hierarchy

Division □ Class □ Order □ Family □ Genus □ Species □ (Subspecies) □ Variety □ Form □ [Cultivar]

Species (basic unit):

Population(s) of plants that are distinct, self perpetuating, usually reproductively isolated.

For laypeople: plants that look “more or less alike”.

Within species: subspecies, varieties, forms (wild) and cultivars (cultivated).

Genus: group of related species (e.g. all oaks = *Quercus*; roses = *Rosa*).

Family: group of related genera (e.g. *Rosaceae*, *Fabaceae*).

Common vs Scientific Names

Common names

- Everyday language; easy to remember.
- Problems:
 - o Same common name for different plants (e.g. “mayflower”).
 - o One plant with many names (e.g. *Erythronium*).
 - o Change between regions (“corn” = maize vs any grain).

Scientific (binomial) names

Introduced by Linnaeus (1753); governed by International Code of Botanical Nomenclature.

- Two parts: Genus + specific epithet
 - o E.g. *Rosa carolina* (pasture rose), *Maianthemum canadense*.
- Latin or Latinized; Genus capitalized, species lower case, both italic/underlined.
- Author name follows species (e.g. *Rosa carolina* L.).
- Intraspecific ranks:
 - o *Rosa carolina* var. *villosa* etc.
 - o Cultivars: non Latin, in quotes, capitalized (e.g. *Pyrus calleryana* ‘Bradford’).

Family names end in -aceae (*Rosaceae*, *Liliaceae*, *Asteraceae*). Some have older alternative names (e.g. *Leguminosae*/*Fabaceae*).

lichen

The word comes from the Ancient Greek word *leikhēn*, which had two main meanings: It referred to the plant-like growth itself, which often appears scaly or crusty on rocks and trees. It also referred to a skin disease characterized by a scaly eruption or ringworm. The name was applied to the organism because of its crust-like, scaly appearance on substrates, resembling certain skin conditions. Lichens are unique because they are a symbiotic association of a fungus and an alga or

Key rule: under one genus, only one correct name per species (earliest valid name).
Understanding check: why are scientific names more reliable than common names?

4. Identifying Plants & Using Keys

Ways to identify plants

- Ask an expert.
- Compare with photos/drawings in field guides.
- Compare with herbarium specimens.
- Use identification keys.

Books can be specialized: food plants, floras, wildflowers, weeds, woody plants, cultivated plants, groups (grasses, orchids, ferns, bryophytes, fungi, algae).

Dichotomous keys

- Series of paired contrasting statements (couplets).
- At each step choose the statement that matches your plant; follow reference to next couplet until you reach a name.
- Often separate keys for family → genus → species.

Hints for using keys (short)

1. Examine the plant first: habit, leaves, flowers, fruits.
2. Use a key specific to your region/habitat.
3. Start at the first couplet; read both choices carefully.
4. Use glossary, lens, and proper measurements.
5. Remember variation within species; if stuck, try both branches and cross check later.
6. Always confirm with description, illustration, and habitat/geography.

Mini question: what is a couplet in a key?

5. Structure of Flowering Plants

Growth forms

- Woody vs herbaceous.
- Woody:
 - o Subshrub – lower stems woody, upper herbaceous.
 - o Shrub – low, several slender trunks.
 - o Tree – taller, few thick trunks + elevated crown.
 - o Woody vine (liana) – climbing/ trailing woody stem.
 - o Leaves: **deciduous** or evergreen.
- Herbaceous:
 - o Annual – 1 growing season.
 - o Winter annual – germinate autumn, flower spring.
 - o Biennial – year 1 vegetative rosette; year 2 flowers, fruits, dies.
 - o Perennial – live >2 years; may have storage roots/stems.
 - o Forbs – broad leaved herbaceous plants.
 - o Succulents – thick, water storing leaves/stems.

5.1 Roots

Functions:

- Absorb water and minerals.
- Anchor plant.
- Storage and asexual reproduction (in some).

deciduous

The word Deciduous comes from the Latin word *dēciduus*, which means "falling off" or "that falls down." It is formed from the Latin verb *dēcidere*, which is a compound of the prefix *dē-* ("down from" or "away") and the verb *cadere* ("to fall"). In botany, it refers to plants (like certain trees and shrubs) that shed their leaves seasonally (usually in autumn) or to any body part that is naturally shed or cast off after a specific period of time.

Types:

- Primary roots □
 - o Taproot – single main vertical root (carrot).
 - o Fibrous roots – many fine roots.
- **Adventitious** roots – arise from stems/leaves.

Enlarged storage roots: carrot, beet, sweet potato, dahlia.

Mycorrhizae ("fungus roots"):

- Fungal hyphae on/in roots.
- Fungi increase water and nutrient uptake; plant supplies carbohydrates.
- Critical for drought and nutrient stress.

5.2 Stems

Functions:

- Support leaves and flowers.
- Conduct water, minerals, sugars (xylem & phloem).
- Can store food, or perform vegetative reproduction.

Features:

- Nodes and internodes; aerial or underground.
- Stolons/runners – surface, horizontal, rooting at nodes.
- **Rhizomes** – underground horizontal stems.
- Tubers – fleshy underground stems (potato).
- Corms – short, swollen stem base (crocus).
- Bulbs – short stem surrounded by fleshy leaf scales (onion, tulip).

These structures often help with asexual reproduction + storage.

Important stem characters (esp. in woody twigs):

- Node, Internode.
- Bud (terminal, axillary), bud scales, bud scale scar.
- Leaf scar with bundle scars.
- Stipule and stipule scars.
- Lenticels (raised pores for gas exchange).

Quick check: how would you distinguish a taproot from a tuber?

6. Leaves

General: lateral appendages; primary photosynthetic organs; extremely variable.

adventitious

The word Adventitious comes from the Latin adjective *adventicius*, which means "coming from abroad," "foreign," or "extrinsic." It is derived from the verb *advenire*, a compound of *ad-* ("to" or "at") and *venire* ("to come"). In botany, it specifically refers to plant parts (like roots, buds, or shoots) that develop in an unusual or unexpected position—that is, they "come to" be where they aren't normally found (e.g., roots growing from a stem or leaf).

mycorrhizae

The word is a compound term derived from Ancient Greek roots, specifically relating to the symbiotic relationship between fungi and plants: *Myco-* (from *mýkēs*), meaning "fungus" or "mushroom." *-rhizae* (from *rhíza*), meaning "root." Therefore, *Mycorrhiza* literally means "fungus-root," accurately describing the intimate, beneficial association between a fungus and the roots of a vascular plant.

rhizomes

It comes from the Greek word (*rhizōma*), which means "mass of roots" or "root-like stalk." It is derived from (*rhíza*), meaning "root" or "base." In botany, it refers to a horizontal underground stem that is distinguished from a true root by the presence of nodes, scale leaves, and buds .

Persistence

- Deciduous – shed during unfavorable season.
- Evergreen – foliage persists >1 year.

Complexity

- Simple leaf – one blade + petiole.
- Compound leaf – blade divided into leaflets.
 - o Palmately compound – leaflets from one point.
 - o Pinnately compound – leaflets along rachis (even bipinnate, tripinnate).

Tip: the axillary bud marks the whole leaf; anything above it (leaflets) is one leaf.

Leaf arrangement on stem

- Alternate – one per node.
- Opposite – two per node.
- Whorled – 3 per node.

Leaf shape (just the key ones)

- Linear, lanceolate, ovate, obovate, elliptic, spatulate, deltoid, orbicular, etc.

Margins

- Entire (smooth), crenate (scalloped), serrate (saw like), dentate (teeth at right angles), lobed (pinnate or palmate).

Tip and base

- Tips: acute, acuminate, obtuse, retuse, emarginate, etc.
- Bases: cuneate, cordate, truncate, oblique, sagittate, etc.

Attachment & venation

- **Petiolate** – with petiole; sessile – no petiole.
- Venation:
 - o Pinnate – midrib + side veins.
 - o Palmate – several primary veins from base.
 - o Parallel – veins nearly parallel.

Surface

- Glabrous – smooth.
- Pubescent – hairy.
- Stellate hairs, glandular hairs, etc.

Special structures

- Tendrils – modified leaf/stem for climbing.
- Thorns – modified stems.
- Spines – modified leaves/parts (incl. stipular spines).
- Prickles – outgrowth of epidermis (e.g. "thorns" on roses).

Understanding check: how would you tell a thorn from a prickle?

7. Flowers & Inflorescences

Flower parts

- Sepals (calyx) – outer, often green.
- Petals (corolla) – colored; together with sepals = perianth; sometimes undifferentiated tepals.
- Stamens (androecium) – filament + anther (pollen).
- Carpels (gynoecium) – stigma, style, ovary (with ovules → seeds).

petiolate

This term comes from the Latin word *petiōlus*, which is the diminutive of *pēs* or *pedis*, meaning "foot." The word literally means "little foot" or "little stalk," and refers to a leaf that possesses a petiole (leaf stalk) that attaches the leaf blade to the stem.

Completeness & sexuality

- Complete – all 4 whorls present (calyx, corolla, androecium, gynoecium).
- Incomplete – one or more missing.
- Perfect (bisexual) – both stamens and carpels.
- Imperfect (unisexual) – staminate (male) or carpellate (female).

Plant condition:

- Synoecious – all flowers perfect.
- Monoecious – male and female flowers on same plant (e.g. maize).
- Dioecious – male and female on different plants.

Inflorescences

Definitions:

- Peduncle – stalk of inflorescence or solitary flower.
- Pedicel – stalk of individual flower.
- Rachis – main axis.
- Bracts, involucre – modified leaves below flowers/heads.

By flowering sequence:

- Determinate – terminal flower opens first (cyme).
- Indeterminate – lower/outer flowers open first; no true terminal (raceme, spike, panicle, umbel, head, etc.).

Key types (indeterminate):

- Raceme – stalked flowers along axis.
- Spike – sessile flowers along axis.
- Panicle – branched raceme.
- Corymb – flat topped; lower pedicels longer.
- Umbel – several stalks from one point; compound umbel has secondary umbels.
- Head (capitulum) – sessile flowers on a common receptacle; typical of Asteraceae (ray + disk florets).

Quick check: which inflorescence type is characteristic of the sunflower family?

8. Fruits, Seeds, Pollination, Dispersal

Fruits

Fruit = ripened ovary, seed = mature ovule.

Dry, dehiscent (split when ripe):

- Follicle (one slit; milkweed).
- Capsule (from compound ovary; several dehiscence types: loculicidal, septicidal, poricidal, etc.).
- Legume (Fabaceae; pod splitting along two sutures).
- Silique/silicle (Brassicaceae).

Dry, indehiscent:

- Achene (one seed, loosely attached; sunflower).
- Nut (large, hard, one seed; acorn).
- Samara (winged achene; elm, ash, maple).
- Schizocarp (splits into one seeded segments).
- Caryopsis (grain; Poaceae – seed coat fused to fruit wall; wheat, maize).

Fleshy:

- Berry – fleshy, many seeds (tomato, grape, banana).
- Drupe – fleshy with stone pit (peach, cherry).
- Pome – fleshy receptacle around core (apple).
- Pepo – hard rind (pumpkin, cucumber).
- Hesperidium – citrus fruits.

Special:

- Aggregate fruit – many ovaries of one flower (raspberry).
- Multiple fruit – ovaries of many flowers fused (pineapple).
- Accessory fruit – non ovary tissue fleshy (strawberry).
- Hip – special rose fruit.

Life cycle: pollination & dispersal

Pollination = transfer of pollen from anther to stigma (often between different plants of same species). Most flowering plants adapted to specific pollinators (insects, birds, bats, etc.) or wind.

Pollinator attraction traits:

- Color, scent (pleasant or foul), nectar guides.
- Rewards: nectar, pollen, food bodies, warmth, breeding sites.
- Flower shape/position matching pollinator behavior.

Wind pollinated flowers: small, unshowy, no scent, often unisexual; produced before leaves.

Seed dispersal = movement of diaspores away from parent.

Why important: reduce competition, colonize new sites, escape pathogens/predators, reduce inbreeding.

Agents:

- Wind – dust seeds, wings (samara), plumes (dandelion), woolly hairs, tumbleweeds.
- Water – floating fruits (coconut), splash cups.
- Animals (external) – hooks, barbs, sticky mucilage (e.g. burdock).
- Animals (internal) – fleshy fruits eaten; seeds defecated; caching by birds/squirrels; ants carrying elaiosome bearing seeds.
- Mechanical – explosive (“ballistic”) fruits (touch me not), shaking in capsules, hygroscopic twisting.

Checkpoint: name one trait useful for wind dispersal and one for animal dispersal.

Seeds and Germination

Seed = embryo + food reserve + seed coat (“baby in a box with its lunch”).

- Need suitable environment: water, oxygen, right temperature, sometimes light/dark.
- Dormancy – viable seed fails to germinate even in favorable conditions.

o Physiological immaturity (after ripening).

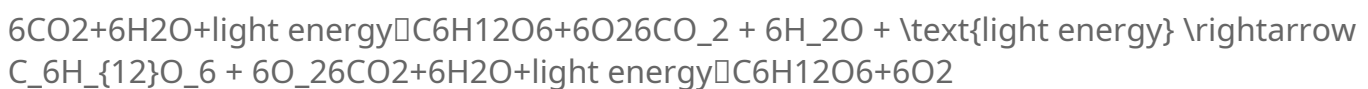
o Hard, impermeable seed coat.

First step: water uptake (imbibition) to start metabolism.

9. Plant Physiology: Photosynthesis, Respiration, Transpiration

Photosynthesis

Overall reaction:



- Occurs in chloroplasts (chlorophyll).
- Light energy splits water; builds sugars.
- Sugars used for respiration or converted to starch, cellulose, proteins, fats, etc.
- Needs: light, CO₂ (through stomata), water, suitable temperature (≈ 65–85 °F for many plants).
- Only a small fraction of light energy is used; rest mainly heats leaves, cooled by transpiration.

Respiration

Controlled oxidation of sugars → energy (ATP), CO₂, water.

- Reverse of photosynthesis in terms of gases.
- Occurs in all living cells, day and night.
- Rate increases with temperature; at high temperatures photosynthesis may stop but respiration continues → uses stored energy, causing stress.

Differences vs photosynthesis (very short):

- Photosynthesis: in chloroplasts, only in light, makes food and stores energy.
- Respiration: in all cells, day and night, uses food and releases energy.

Transpiration

Loss of water vapor mainly through leaf stomata.

Roles:

- Pulls water and minerals from roots.
- Cools leaves (evaporative cooling).
- Helps move sugars and chemicals.
- Maintains turgor.

Rate increases with higher temperature, wind, and lower humidity.

Understanding check: why does closing stomata during drought slow both transpiration and photosynthesis?

10. Transport: Xylem vs Phloem

Xylem

- Conducts water + some solutes from roots → shoots (unidirectional).
- Water enters roots by osmosis (water moves into cells with more solutes).
- Root pressure + transpirational pull create continuous water column from roots to leaves.
- Main conducting cells: tracheids and vessels (dead, hollow).

Phloem

- Conducts mainly sugars (plus amino acids, hormones, minerals) from sources (photosynthesizing leaves, storage organs) to sinks (growing tips, roots, fruits, seeds).
- Direction can be up or down, depending on source–sink relations.
- Main cells: sieve tubes (living).

Mobile nutrients can move via phloem from older to younger tissues; deficiency symptoms appear first in old leaves.

Quick check: one key difference in direction of transport between xylem and phloem?

11. Growth, Meristems, Bark, Hormones

Growth vs Development

- Growth – irreversible increase in size/weight (cell division + enlargement).
- Development – differentiation of cells/tissues/organs; changes like flowering.

Meristems

- Apical meristems (root/shoot tips) → primary growth (length).
 - o Indeterminate growth (continuous) vs determinate (stops after flowering/fruiting).
- Lateral meristems:
 - o Vascular cambium – adds secondary xylem (wood) and secondary phloem.
 - o Cork cambium – forms cork + phelloderm = periderm (outer protective tissue).
- Secondary growth thickens stems/roots in woody plants.

Bark

- All tissues outside vascular cambium.
 - o Inner bark – living tissue, mainly phloem + inner periderm.
 - o Outer bark – dead tissue outside innermost cork cambium.
- Lenticels in periderm allow gas exchange.

Plant hormones (just two examples)

- Auxin – produced in young leaves and seeds.
 - o Roles: phototropism, gravitropism, root induction, fruit development, apical dominance.
- Absciscic acid (ABA) – produced in mature leaves under water stress.
 - o Roles: stomatal closure, transport of food, seed and bud dormancy.